

Airbnb (Italy) Power BI Dashboard Report

Listings, Pricing, and Host Performance Insights

Prepared on: 19 Feb 2026

This report explains the design, metrics, and key insights from three Power BI dashboards built on an Airbnb dataset for Italian cities.

Problem Statement:

Stakeholders (hosts, property managers, and analysts) lack a consolidated view to benchmark Airbnb performance across Italian cities. Without a central dashboard, it is hard to determine where supply is concentrated, which property types drive the market, what typical price ranges look like, and how guest ratings and Superhost status relate to pricing. The goal of this project is to build a set of Power BI dashboards that translate the dataset into actionable insights for market comparison, pricing decisions, and host quality improvement.

1. Executive Summary

The Power BI report is designed to help stakeholders understand Airbnb supply and performance across Italy through three complementary lenses:

- Dashboard 1: Market overview (listings volume, reviews, ratings, average price) with city and neighbourhood drill-down.
- Dashboard 2: Pricing insights (distribution, pricing by property type, relationship between price and ratings, and superhost pricing).
- Dashboard 3: Host insights (host vs superhost contribution, rating differences, trends over time).

Key takeaways visible from the dashboards:

- Firenze (example filter shown) has 13.58K listings and ~3M reviews with strong guest sentiment (avg location score 4.80; value-for-money 4.66).
- Supply in Firenze is dominated by Entire home listings (11,240) followed by Private room (2,252); Hotel room (103) and Shared room (34) are minimal.
- Centro Storico contributes the majority of Firenze's listings (9,862 of 13,583), indicating a strong concentration of supply in the historic core.
- Pricing is right-skewed: most listings cluster at lower price bins with a long tail of premium outliers.
- Superhost listings achieve higher average rating (about 4.9) than non-superhost hosts (about 4.6), but average prices between the two groups are broadly similar.

2. Dataset and Metric Definitions

The report uses an Airbnb listings dataset for Italian cities (e.g., Firenze, Milano, Napoli, Roma, Venezia). The dashboards are built to support interactive exploration through slicers for City, Neighbourhood, Property type, Season, and Price (or Price bins).

2.1 Core KPIs used across pages

KPI	Meaning (business interpretation)
Count / Num of Listings	Number of active listings in scope of the applied filters.
Total reviews	Total review count (proxy for demand and booking volume over time).
Average Rating score	Average guest rating across listings in scope (0–5 scale).
Avg Location score	Average location sub-score (how guests rate location).
Avg Value for money	Average value-for-money sub-score (perceived value relative to price).
Average of Price	Average nightly price in scope (in EUR).
Host Listing / Superhost Listings	Counts attributed to host category; supports mix and performance comparisons.

2.2 How to read filters (slicers)

Slicers on the right panel control the scope of every visual on the page. A typical workflow is: choose City → optionally narrow to Neighbourhood → select Property type → inspect Season and Price patterns. If you want to compare across cities, select multiple cities (Dashboards 2 and 3 support multi-select).

3. Dashboard 1 — Airbnb Italian Listings Overview

Purpose: provide a fast market overview of listing supply, review volume, guest sentiment, and price levels, with drill-down to neighbourhoods and seasonality.

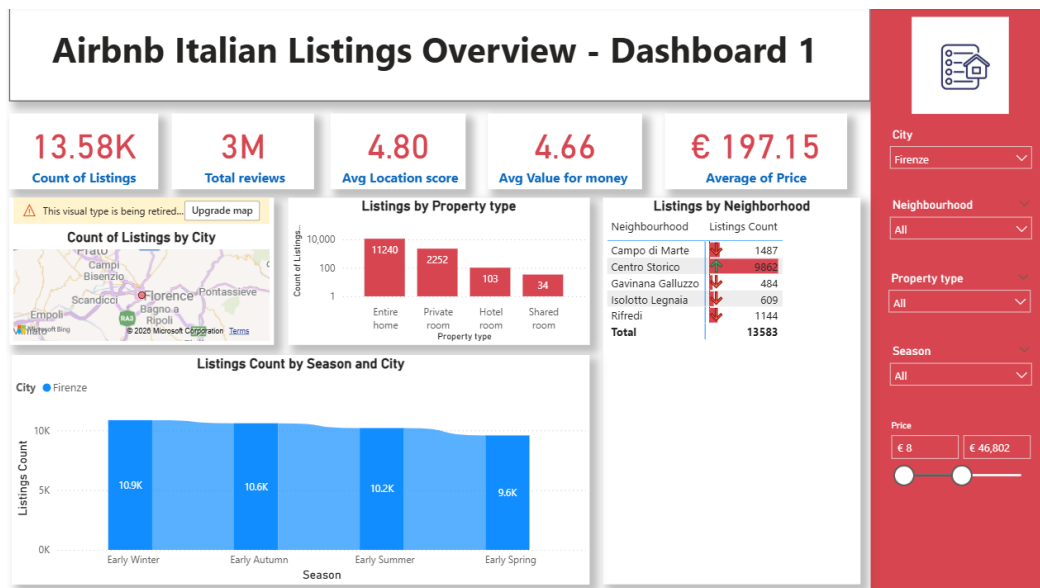


Figure 1. Dashboard 1 (example filter: City = Firenze).

3.1 What the page shows

- Top KPI cards summarize the overall market for the selected filters (listings, total reviews, location score, value-for-money score, average price).
- A map visual supports geographic context and quick spotting of city-level listing concentration.
- Bar chart: listings by property type reveals the accommodation mix (entire homes vs rooms).
- Table: listings by neighbourhood highlights where supply is concentrated and enables quick ranking.
- Season chart: compares listing counts across seasons to indicate seasonality patterns.

3.2 Key metrics (as shown in the screenshot)

Metric (Firenze selection)	Value
Count of Listings	13.58K (13,583)
Total reviews	3M
Avg Location score	4.80
Avg Value for money	4.66
Average of Price	€197.15

3.3 Insights (Firenze example)

- Supply concentration: Centro Storico alone contributes 9,862 listings ($\approx 72.6\%$ of the Firenze total 13,583), indicating the historic center is the dominant supply cluster.

- Property mix: Entire home listings (11,240) overwhelmingly dominate, suggesting higher prevalence of whole-property stays vs shared accommodation.
- Long-tail neighbourhoods: areas like Gavinana Galluzzo (484) and Isolotto Legnaia (609) are meaningful but much smaller than the center, supporting targeted neighbourhood-level positioning.
- Seasonality: listing counts are highest in Early Winter (~10.9K) and trend down to Early Spring (~9.6K), a pattern that may reflect availability/listing activity rather than demand.

3.4 Suggested business actions enabled by this page

- If you are a host or property manager: benchmark your neighbourhood against the market concentration and consider differentiation if you are in a highly saturated area (e.g., Centro Storico).
- If you are analysing pricing strategy: use the property type mix first, then drill into neighbourhoods to identify where higher prices may be supported.
- If you are planning seasonal campaigns: use the season view to identify when listing availability changes and align marketing/inventory operations accordingly.

4. Dashboard 2 — Airbnb Listings Pricing Insights

Purpose: explain how prices vary by property type, how prices are distributed, whether ratings relate to price, and whether superhosts command a price premium.

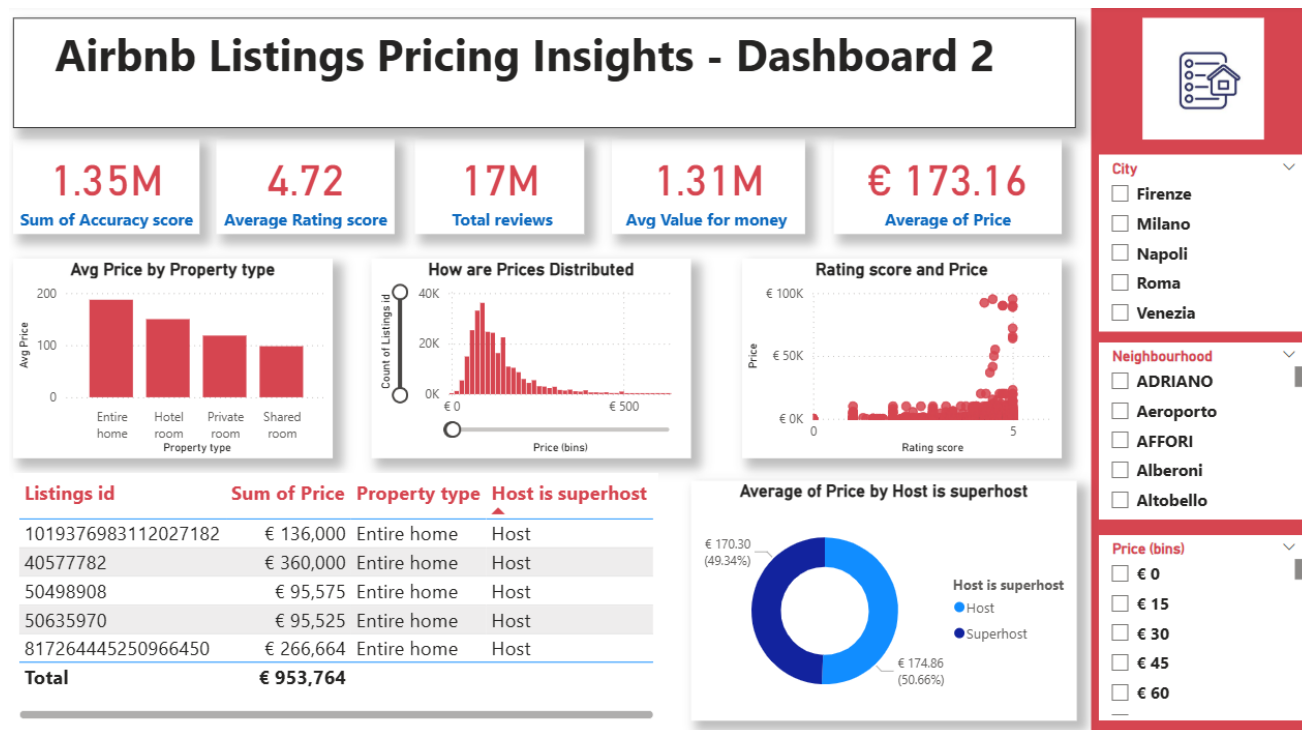


Figure 2. Dashboard 2 (Pricing insights across selected cities/neighbourhoods).

4.1 What the page shows

- KPI cards summarize score/reviews/average price for the current selection.
- Avg Price by Property type identifies which accommodation types typically price higher.
- Histogram shows the price distribution and highlights skewness/outliers.

- Scatter plot shows the relationship between rating score and price (e.g., whether higher-rated listings charge more).
- Donut chart compares average price between Superhost vs Host segments.

4.2 Key metrics (as shown in the screenshot)

Metric	Value
Sum of Accuracy score	1.35M
Average Rating score	4.72
Total reviews	17M
Value-for-money (displayed)	1.31M
Average of Price	€173.16
Example table total (Sum of Price)	€953,764

4.3 Insights

- Typical pricing by accommodation type: Entire homes show the highest average price, while shared rooms are lowest, reflecting privacy and capacity differences.
- Distribution shape: the histogram is strongly right-skewed (many listings at lower price bins) with a long premium tail, indicating a small number of very expensive listings that can distort averages.
- Ratings vs price: most listings cluster at high ratings (around 4–5), while price varies widely; this suggests ratings alone do not fully explain pricing and other factors (location, property type, amenities) are key.
- Superhost price effect: the average price for Superhosts (~€174.86) is slightly higher than Hosts (~€170.30), but the difference is small, implying Superhost status is more strongly linked to service quality than to large price premiums.

4.4 Suggested business actions enabled by this page

- When reporting average price, always cross-check with the distribution/histogram to ensure outliers are not driving conclusions.
- For pricing optimization, segment first by property type and neighbourhood; then validate whether higher prices correspond to stronger ratings within each segment.
- Use the superhost comparison to position quality improvements as a path to better conversion, not only as a way to justify major price increases.

5. Dashboard 3 — Airbnb Listings Host Insights

Purpose: compare Host vs Superhost segments on volume, ratings, pricing, and how the listing base evolves over time.

Airbnb Listings Host Insights - Dashboard 3

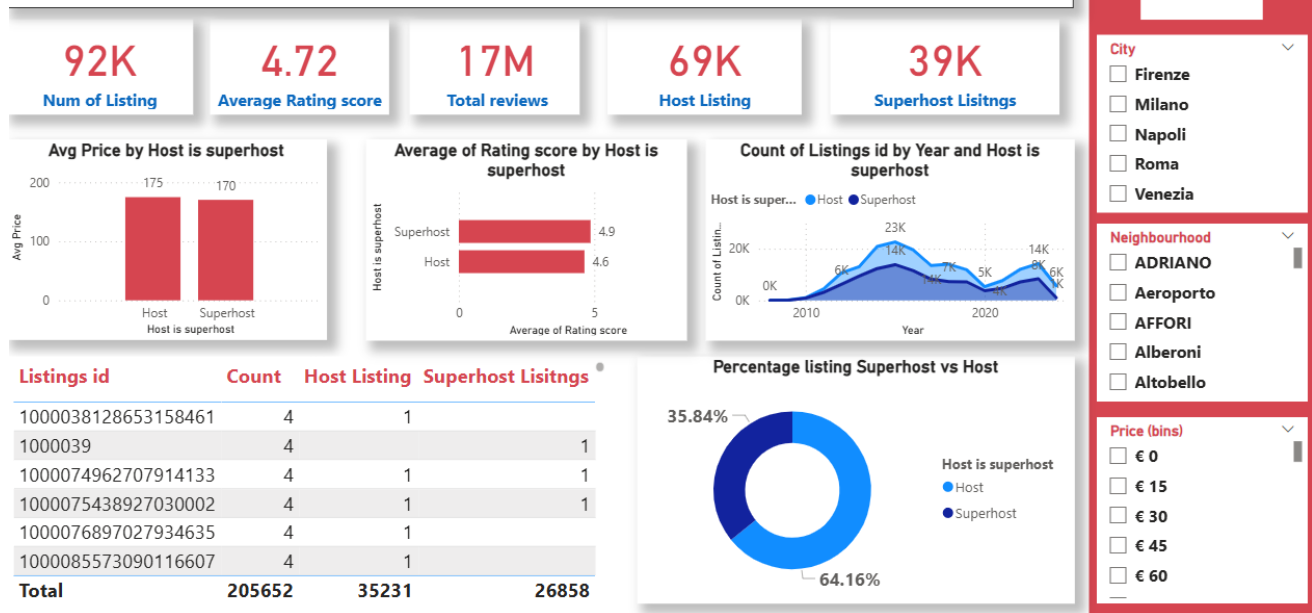


Figure 3. Dashboard 3 (Host vs Superhost insights).

5.1 What the page shows

- KPI cards summarize the listing base and review volume, plus host and superhost counts for the current selection.
- Bar chart compares average price for Hosts vs Superhosts.
- Bar chart compares average rating for Hosts vs Superhosts.
- Time series shows how listing volume changes over the years and how Host vs Superhost contributions evolve.
- Donut chart shows the percentage split between Host and Superhost listings.

5.2 Key metrics (as shown in the screenshot)

Metric	Value
Num of listing	92K
Average Rating score	4.72
Total reviews	17M
Host Listing	69K
Superhost Listings	39K
Host vs Superhost share (donut)	Host 64.16% Superhost 35.84%
Avg rating by segment	Superhost ~4.9 Host ~4.6

5.3 Insights

- Host mix: Hosts represent ~64% of listings while Superhosts represent ~36%, meaning Superhosts are a substantial minority and not a niche segment.
- Quality signal: Superhosts score materially higher on average rating (~4.9 vs ~4.6), reinforcing Superhost status as a strong quality indicator.
- Pricing parity: the average price bars are close between Hosts and Superhosts, implying quality leadership is not solely achieved through charging higher prices.
- Growth pattern: the time series shows a clear rise through the mid-2010s with a peak around 2016, followed by a decline and then a smaller rebound; this supports time-based analysis (market expansion, policy changes, or shocks).

5.4 Suggested business actions enabled by this page

- If you are managing host quality: use the rating gap between Superhosts and Hosts to prioritize service/operations improvements that lift the baseline Host experience.
- If you are analysing supply growth: use the time series to frame period-based analysis (pre-peak vs post-peak), then compare cities/neighbourhoods to isolate drivers.
- If you are designing an acquisition strategy: target neighbourhoods/cities where Superhost share is low but demand (reviews) is high — indicating quality upgrade opportunity.

6. Conclusion and Recommendations

Together, the three dashboards provide a complete story: market supply and sentiment (Dashboard 1), pricing behavior and outliers (Dashboard 2), and host segment performance and evolution (Dashboard 3).

- Always interpret price using both averages and distribution to avoid outlier bias.
- Use neighbourhood concentration views to guide saturation-aware strategy (differentiation in dense cores, growth in under-served areas).
- Treat Superhost status as a quality lever rather than a pure pricing lever; focus on experience improvements to raise ratings.
- Add a consistent time-period filter across all pages (if not already) to support clear before/after comparisons.

Appendix A — Dashboard Interaction Tips

- Hover tooltips for exact values; click a bar/segment to cross-filter all visuals on the page.
- Use Ctrl-click for multi-select in charts (Power BI default behaviour).
- Reset filters using the eraser icon on slicers (or a page-level reset button if you add one).
- When comparing cities, keep price bins consistent to ensure fair distribution comparisons.